

PYTHON SOCKET PROGRAMMING NOTES

Basics of Socket Programming with Real Life Examples

Learn Client and Server Communication in Python

Introduction to Socket Programming

Socket programming ka use do devices ke darmiyan communication ke liye hota ha.

Python sockets ki madad se network connection establish karta ha.

Ye client aur server model par kaam karta ha.

Sockets internet aur local networks dono mein use hote hain.

Developers chat apps aur multiplayer games mein sockets use karte hain.

Socket programming real time communication provide karti ha.

Python socket module ke through networking easy ban jati ha.

Ye cybersecurity aur networking mein bohat important topic ha.

Beginners ko basic networking concepts samajhna chahiye.

Socket programming modern internet applications ki base ha.

Real Life Example: Jaise do log mobile call par baat karte hain waise hi sockets devices ko connect karti hain.

Client and Server

Socket programming mein do important parts hote hain client aur server.

Server data provide karta ha aur client request bhejta ha.

Client server se connect ho kar communication karta ha.

Server multiple clients ko handle kar sakta ha.

Python mein alag alag programs client aur server ke liye likhe jate hain.

Websites bhi client server architecture use karti hain.

Ye networking ka basic aur important concept ha.

Server continuously requests ka wait karta ha.

Client sirf jab zaroorat ho tab request bhejta ha.

Har network application mein client aur server ka role hota ha.

Real Life Example: Restaurant mein customer client hota ha aur waiter/server service provide karta ha.

Creating a Socket

Python mein socket create karne ke liye socket() function use hota ha.
Socket communication start karne ka pehla step hota ha.
Socket module ko pehle import kiya jata ha.
Sockets different protocols support karte hain.
TCP aur UDP common socket types hain.
Socket object network communication manage karta ha.
Developers applications ke liye custom connections bana sakte hain.
Socket creation networking programming ka basic ha.
Python socket syntax beginners ke liye easy hoti ha.
Ye network applications ki foundation hoti ha.

Real Life Example: Jaise ghar banane se pehle darwaza lagaya jata ha waise hi communication ke liye socket create ki jati ha.

TCP Protocol

TCP ek reliable communication protocol ha.
Ye data ko proper order mein transfer karta ha.
TCP errors ko detect aur fix karne mein help karta ha.
Socket programming mein TCP bohat use hota ha.
Websites aur banking systems TCP use karte hain.
Python SOCK_STREAM ke through TCP sockets create karta ha.
TCP secure aur stable communication provide karta ha.
Ye large data transfer ke liye suitable hota ha.
Networking mein TCP ka bohat important role ha.
Beginners ko TCP basics zaroor seekhne chahiye.

Real Life Example: Courier service jo parcel safely deliver karti ha wo TCP ki tarah hoti ha.

Sending and Receiving Data

Sockets ke through data send aur receive kiya jata ha.
Python send() aur recv() methods provide karta ha.
Client aur server messages exchange karte hain.
Data bytes ki form mein transfer hota ha.
Chat applications isi concept par kaam karti hain.
Developers live communication systems create karte hain.
Receiving data ke liye server continuously listen karta ha.
Python encoding aur decoding bhi use karta ha.
Ye real time networking ka important feature ha.
Socket communication internet applications ka major part ha.

Real Life Example: WhatsApp par messages bhejna aur receive karna socket communication jaisa hota ha.

Closing a Socket

Communication complete hone ke baad socket close karna zaroori hota ha.
Close() method network connection ko terminate karta ha.
Agar socket close na ho to resources waste ho sakte hain.
Professional programmers always sockets properly close karte hain.
Socket closing secure communication mein help karti ha.
Ye system performance ko better banata ha.
Network errors ko avoid karne ke liye closing zaroori ha.
Python safe networking practices ko support karta ha.
Socket close karna good programming habit ha.
Har networking application mein closing important step hota ha.

Real Life Example: Phone call complete hone ke baad call disconnect karna socket close karne jaisa ha.
